**KENYA INSTITUTE OF CURRICULUM DEVELOPMENT**
*A Skilled and Ethical Society*

# SENIOR SCHOOL CURRICULUM DESIGN

# CORE MATHEMATICS

## GRADE 10

**2025**

DRAFT

First Published in 2025

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## FOREWORD

The Government of Kenya (GoK) is committed to ensuring that policy objectives for Education, Training and Research meet the aspirations of the Kenya Constitution 2010, the Kenya Vision 2030, National Curriculum Policy 2019, the United Nations Sustainable Development Goals (SDGs) and the Regional and Global conventions to which Kenya is a signatory.

Towards achieving the mission of Basic Education, the Ministry of Education (MoE) embarked on a reform process that culminated in the adoption of a Competency Based Education (CBE) system in 2017. This came after extensive research, stakeholder engagement international benchmarking and consideration of the 21st Century socio-economic realities. CBE in Kenya adopted a 2-6-3-3 structure where the numbers indicate the years learners take at various levels in Basic Education (Pre-Primary, Primary, Junior School and Senior School; respectively). CBE has for critical elements: The Competency Based Curriculum (CBC), Pedagogy (teaching and learning), Resources and the Competency Based Assessment. The MoE oversees the provision of resources while the other components (Curriculum, Pedagogy and Assessment are handled by the Kenya Institute of Curriculum Development, (KICD), Teachers Service Commission, (TSC) and the Kenya National Examinations Council (KNEC); respectively.

KICD has successfully developed curriculum designs that have been used in the implementation of the Competency Based Curriculum (CBC) for the Pre-Primary, Primary and Juniors School levels. The first cohort of learners pursuing the CBC will transit to Senior School (Grades 10-12) in 2026. The curriculum designs at this level are expected to facilitate learners to acquire relevant knowledge, skills, values and attitudes (competencies) in preparation for tertiary education and the world of work. I urge all agencies of the GoK and other stakeholders in Education to use the designs to plan for effective and efficient implementation of Competency Based Education.

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**MR. JULIUS MIGOS OGAMBA**
**CABINET SECRETARY,**
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## PREFACE

The Ministry of Education (MoE) through the State Department for Basic Education, has been implementing the Competency Based Education (CBE) since the adoption of the *Basic Education Curriculum Framework* (KICD, 2017) during the national convention in January 2017. The Kenya Institute of Curriculum Development (KICD) developed Competency Based Curriculum (CBC) designs that were used for pilot studies in 2017 and 2018 before the roll-out at Pre-Primary and Primary Early Year (Grades 1 to 3 in 2019. The curriculum was then introduced in the subsequent grades through *a phase in –phase out* strategy, to ensure a smooth transition. Currently, the first cohort of CBC learners are at Grade 9 – the last grade at Junior School. The implementation of CBE has since benefited immensely from the recommendations of the *Presidential Working Party on Education Reforms* (PWPER) that were released in August, 2023 and fully implemented, in terms of curriculum designs.

Grade 10 is the entry class of the Senior School (SS) in Basic Education, a very critical level where learners shall have the opportunity to focus on pathways, tracks and subjects of their choices guided by the potentials demonstrated at earlier levels. The Senior School shall; therefore, form a foundation for further education and training and/or initial competencies for the Industry. This is an important step in the realisation of the Vision and Mission of the curriculum reforms as enshrined in the Sessional Paper No. I of 2019 whose title is: *Towards Realizing Quality, Relevant and Inclusive Education and Training for Sustainable Development* in Kenya.

The Grade 10 curriculum designs, like the designs for earlier levels, are anchored on the Kenyan National Goals of Education. They also present essence statements, general and specific expected learning outcomes as well as strands and sub strands for the learning areas. The designs also outline suggested learning experiences, key inquiry questions, core competencies, Pertinent and Contemporary Issues (PCIs), values and suggested assessment.

The Ministry of Education, through the State Department for Basic Education commits to provide resources and relevant guidelines for a smooth implementation of the CBC. It is expected that the designs will guide teachers to effectively facilitate learners to attain the expected learning outcomes for Grade 10 specifically and Senior School in general.

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## ACKNOWLEDGEMENT

The Kenya Institute of Curriculum Development (KICD) Act Number 4 of 2013 (Revised 2019) mandates the Institute to develop curricula, programmes and curriculum support materials for Basic and Tertiary Education and Training. The curriculum development process involves research, international benchmarking and robust stakeholder engagement. Through a systematic and consultative process, the KICD conceptualised the Competency Based Curriculum (CBC) as captured in the *Basic Education Curriculum Framework* (KICD,2017), that responds to the demands of the 21st Century, the United Nations Sustainable Development Goals (SDGs). African Agenda 2063, East African Community Protocol and the aspirations captured in the Kenya Constitution 2010 and the Kenya Vision 2030.

KICD receives its funding from the GoK to enable the successful achievement of the stipulated mandate and implementation of the Government and Education sector plans. The Institute also receives support from development partners targeting specific programmes. The Grade 10 curriculum designs were developed with the support of the World Bank and GoK through the *Kenya Primary Education Equity in Learning Programme* (KPEELP); a project coordinated by the MoE. Therefore, the Institute is very grateful to the GoK, through the MoE and the development partners (especially The World Bank) for support in terms of policy guidance, resource provision and capacity building. Specifically, I wish to express special thanks to the Cabinet Secretary-MoE and the Principal Secretary-State Department for Basic Education.

We also wish to acknowledge the KICD curriculum development officers and other staff, all teachers, educators who took part as panellists; the institutions that participated and representatives of various stakeholders for their roles in the development of the Grade 10 curriculum designs. In relation to this, we acknowledge the support of the Chief Executive Officers of the Teachers Service Commission (TSC), the Kenya National Examinations Council (KNEC) and other agencies in the sector for their support in the process of developing these designs.

Finally, we are very grateful to the KICD Council Chairperson and other members of the Council for consistent guidance and oversight. We assure all teachers, parents and other stakeholders that these curriculum designs will effectively guide the implementation of the CBE at Grade 10 and preparation of learners for transition to further education and training or the world of work.

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|  |  |  | **Project**<br>Make a simple clinometer using locally available materials and use it as a learning resource |  |

**Core Competencies to be developed:**

- Creativity and Imagination: the learner makes a simple clinometer using locally available material and uses it as a learning resource
- Critical Thinking and Problem Solving: the learner discusses and makes observations about the sine and cosine of complementary angles, and applies trigonometric ratios in the solution of triangles
- Digital literacy: the learner uses digital devices and other resources such as books, manuals, and journals to learn more about trigonometric ratios

**Values:**

- Responsibility: the learner takes care of mathematical tables and calculators used in determining trigonometric ratios
- Unity: the learner works in a group, for instance, when using an isosceles right-angled triangle and an equilateral triangle to generate the trigonometric ratios of special angles 30°, 45°, 60°, and 90°

**Pertinent and Contemporary Issues (PCIs):**

- Environmental Education and Protection: the learner identifies shapes in the immediate environment that have right-angled triangles, and makes a simple clinometer using materials from the immediate environment
- Parental Engagement and Empowerment: the learner gets support from parents/guardians to get resources to make a clinometer.

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**Sub-Strand 5: Area of Polygons**

| Strand | Sub-Strand | Specific Learning Outcomes | Suggested Learning Experiences | Suggested Key Inquiry Question(s) |
|---|---|---|---|---|
| **2.0** | **2.5 Area of** | By the end of the sub-strand | The learner is guided to: | 1. How do we |

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| **Measurements and Geometry** | **Polygons**<br>- *Area of a triangle using* $Area = \frac{1}{2}abSinC$<br>- *Area of a triangle using Heron's formula*<br>- *Area of quadrilaterals*<br>- *Area of regular and irregular polygons*<br>- *Application of area of polygons* | the learner should be able to:<br>a) derive the formula for the area of a triangle given two sides and an included angle,<br>b) work out the area of a triangle given two sides and an included angle,<br>c) determine the area of a triangle using Heron's formula,<br>d) determine the area of quadrilaterals in different situations,<br>e) work out the area of regular heptagon and octagon,<br>f) determine the area of irregular polygons in different situations, | - discuss in groups and use trigonometric ratios to generate the formula for the area of a triangle given two sides and an included angle. For example, $Area = \frac{1}{2}abSinC$<br>- use the formula $Area = \frac{1}{2}abSinC$ to calculate the area of different triangular shapes,<br>- use the Heron's Formula (Hero's Formula) to work out the areas of various triangles,<br>- identify different quadrilaterals and work out their areas,<br>- identify from the immediate environment objects with the shape of regular polygons and calculate the area of regular polygons up to heptagon, | work out the area of polygons?<br>2. How do we apply the concept of the area of polygons in real-life situations? |

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| | | g) explore the area of polygons as used in real-life situations. | - use objects from the immediate environment that have shapes of irregular polygons, discuss and work out their area,<br>- research and discuss in a group the use of the area of polygons in real-life situations. | |

**Core Competencies to be developed:**

- Critical Thinking and Problem Solving: the learner uses trigonometric ratios to generate the formula for a triangle given two sides and an included angle.
- Creativity and Imagination: the learner identifies from the immediate environment objects in the shape of regular polygons and calculates their area.
- Learning to Learn: the learners research and discuss the use of the area of polygons in real-life situations.

**Values:**

- Patriotism: the learner uses the objects from the environment and takes care of the environment.
- Responsibility: the learner identifies objects from the immediate environment to use in calculating the area of regular polygons.
- Peace: the learner harmoniously discusses in groups and uses trigonometric ratios to generate the formula for the area of a triangle given two sides and an included angle.

**Pertinent and Contemporary Issues (PCIs):**

Environmental Education: the learner collects objects from the immediate environment to be used as learning materials for area of polygons.

**Sub-Strand 6: Area of a Part of a Circle**

| **Strand** | **Sub-Strand** | **Specific Learning Outcomes** | **Suggested Learning Experiences** | **Suggested Key Inquiry Question(s)** |
|---|---|---|---|---|
| **2.0 Measurements and Geometry** | **2.6 Area of a Part of a Circle**<br>(12 lessons)<br>- *Area of an annulus*<br>- *Area of a sector of a circle*<br>- *Area of an annular sector*<br>- *Area of a segment*<br>- *Area of* | By the end of the sub-strand the learner should be able to:<br>a) determine the area of an annulus in different situations,<br>b) work out the area of a sector of a circle in real-life situations,<br>c) determine the area of an annular sector in different | The learner is guided to:<br>- use circular shapes or objects to identify concentric rings formed by inner and outer space. Hence work out the area of an annulus,<br>- work in a group, and use paper cutouts to make sectors of circles to determine their areas as part of a circle,<br>- use different examples to illustrate and work out the area of an annular sector, for example area covered by a car wiper blade when cleaning the windscreen, | How do we use the concept of the area of a part of a circle in real life? |

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| | *common region between two intersecting circles*<br>• *Application on the use of the* | situations,<br>d) work out the area of a segment of a circle,<br>e) determine the area of common region between two | • draw different segments of a circle and calculate their area,<br>• draw illustrations of two intersecting circles to identify the common area/part between the two circles. Work out the area of common area/part. | |

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| | *area of a part of a circle* | intersecting circles,<br>f) apply the area of a part of a circle in real-life situations,<br>g) explore the use of the area of a part of a circle in real-life situations. | • relate and work out the area of a part of a circle in real-life situations,<br>• use digital devices and other resources to learn more about the area of a part of a circle.<br>**Project**<br>• make a dartboard of different numbers of concentric circles from locally available materials. Paint different colours on the annular sectors and the concentric circles. Discuss and create rules for playing and scoring the game.<br>• make necklaces using beading techniques of different coloured annular sectors. | |

**Core Competencies to be developed:**

- Creativity and Imagination: the learner makes an improvised dartboard of different numbers of concentric circles.
- Learning to Learn: the learner draws illustrations of two intersecting circles to identify the common area/part between the two circles.
- Communication and collaboration: the learners engage to improvise locally available materials to make dartboards and beaded necklaces.

**Values:**

- Responsibility: the learner makes dartboard of different numbers of concentric circles from the immediate environment
- Unity: the learner collaborates in a group and uses paper cutouts to make sectors of circles to determine their areas as part of a circle

**Pertinent and Contemporary Issues (PCIs):**

- Parental Engagement and Empowerment: the learner engages parents in using locally available materials for making improvised beaded necklaces.
- Environmental Education: the learner sources locally available materials to make necklaces and dartboards.
- Learner Support Programmes (Peer Education): the learner works with others to make necklaces using beading techniques of different coloured annular sectors.

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**Sub-Strand 7: Surface Area and Volume of Solids**

| Strand | Sub-Strand | Specific Learning Outcomes | Suggested Learning Experiences | Suggested Key Inquiry Question(s) |
|---|---|---|---|---|
| **2.0 Measurements and Geometry** | **2.7 Surface Area and Volume of Solids**<br>• *Surface Area of Solids*<br>• *Volume of Solids* | By the end of the sub-strand, the learner should be able to:<br>a) determine the surface area of prisms, pyramids, cones, frustums and spheres,<br>b) calculate the surface area of composite solids,<br>c) calculate the volume of prisms, pyramids, cones, frustums and spheres,<br>d) determine the volume of composite solids,<br>e) explore the use of the surface area and volume of solids in real-life situations. | The learner is guided to:<br>• use models and nets of prisms, pyramids, cones, frustums, and spheres to calculate surface areas.<br>• identify the components of a composite solid, calculate the surface area of each component and obtain the total surface area of the composite solid.<br>• use models of a cone and a cylinder of the same height and base area and an appropriate substance such as sand or water, to demonstrate that the volume of the cone is a third of the volume of the cylinder.<br>• discuss in a group and work out the volume of prisms, pyramids, cones, frustums, and spheres.<br>• discuss in a group the components of composite solids, work out the volume of the components and obtain the total volume.<br>• use appropriate containers from the local environment to work out the volume and capacity of the prism, pyramid, cone, frustum and sphere.<br>• use digital devices and other resources to work out the surface area and volume of solids. | 1. How do we determine the surface area and volume of solids?<br>2. Why do we determine the surface area and volume of solids? |

**Core Competencies to be developed:**
- Critical Thinking and Problem Solving: the learner uses a cylinder and a cone of the same radius and height, and a substance such as sand or water, to determine the volume of a cone as a third of the volume of a cylinder.
- Creativity and Imagination: the learner uses a cylinder and a cone of the same radius and height, and a substance such as sand or water, to determine the volume of a cone as a third the volume of a cylinder.

**Values:**
- Unity: the learner discusses in a group and establishes the relationship between the volume of a cone and the volume of a cylinder.
- Respect: the learner listens to others' opinions as they discuss in groups and work out the volumes of prisms, pyramids, cones, frustums and spheres.

**Pertinent and Contemporary Issues (PCIs):**
Environment and technology: as the learner collects and uses containers from the immediate environment and uses them to discuss and work out their volumes.

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**Sub-Strand 8: Vectors I**

| **Strand** | **Sub-Strand** | **Specific Learning Outcomes** | **Suggested Learning Experiences** | **Suggested Key Inquiry Question(s)** |
|---|---|---|---|---|
| **2.0 Measurements and Geometry** | **2.8 Vectors I**<br>- *Vector and scalar quantities.*<br>- *Vector notation*<br>- *Representation of vectors*<br>- *Equivalent vectors*<br>- *Addition of vectors*<br>- *Multiplication of vectors by scalar*<br>- *Column vectors*<br>- *Position vectors*<br>- *Magnitude of a vector*<br>- *Midpoint of a vector*<br>- *Translation vector* | By the end of the sub-strand the learner should be able to:<br>a) explain vector and scalar quantities with illustrations<br>b) use vector notations in different ways<br>c) represent vectors geometrically in different situations<br>d) identify equivalent vectors in different situations<br>e) add vectors in different situations<br>f) multiply vectors by scalars<br>g) determine column vectors in different situations<br>h) determine position vectors in different situations<br>i) work out magnitude of a vector in different situations<br>j) determine the midpoint of a vector in different situations<br>k) determine translation vector as a transformation<br>l) appreciate the use of vectors in real-life situations | The learner is guided to:<br>- search and brainstorm on the meaning of the terms vector and scalar quantities and illustrate giving examples,<br>- practise writing vector notations on charts<br>- illustrate vectors on charts, grids and regular plane figures,<br>- discuss the meaning of arrows and lines on roads and road signs in relation to movement on the roads to enhance safety,<br>- demonstrate equivalent vectors on charts, grids and Cartesian coordinates,<br>- illustrate addition of vectors using charts and grids to get the resultant vector,<br>- illustrate multiplication of vectors by a scalar on charts and grids,<br>- express vectors in column form on Cartesian planes, grids and charts,<br>- discuss and illustrate position vectors using any arbitrary reference points like the origin,<br>- work out magnitude of different vectors,<br>- find the midpoint of different vectors,<br>- use vectors to illustrate translation as a transformation,<br>- use digital devices to watch videos and play games involving vectors. | 1. What are vectors?<br>2. Where are vectors used in real-life situations? |

**Core competencies to be developed:**

- Learning to learn: As the learner uses charts and grids to practise addition of vectors.
- Self-efficacy: As the learner shares and uses resources from the immediate environment such as plane figures or flip charts to demonstrate equivalent vectors.

**Values:**

- Responsibility: As the learner uses resources from the immediate environment such as plane figures, flip charts and relate equivalent vectors.
- Unity: As the learner searches and brainstorms on the meaning of the terms vector and scalar quantities and illustrates giving examples.

**Pertinent and Contemporary Issues (PCIs):**

- Education for Sustainable Development - Environmental education as learners practice addition of vectors using charts and grids made from resources in the immediate environment.
- Road safety: as the learner discusses the meaning of arrows and lines on roads and road signs in relation to movement on the roads to enhance safety.

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**Sub-Strand 9: Linear Motion**

| Strand | Sub-Strand | Specific Learning Outcomes | Suggested Learning Experiences | Suggested Key Inquiry Question(s) |
|---|---|---|---|---|
| **2.0 Measurements and Geometry** | **2.9 Linear Motion**<br>- *Distance, displacement, speed, velocity, and acceleration*<br>- *Velocity and acceleration*<br>- *Displacement time graph*<br>- *Interpretation of displacement-time graph*<br>- *Velocity time graph*<br>- *Interpretation of velocity - time graph* | By the end of the sub-strand, the learner should be able to:<br>a) explain the terms distance, displacement, speed, velocity and acceleration in real-life situations,<br>b) determine velocity and acceleration in different situations,<br>c) draw a displacement-time graph of different situations,<br>d) interpret displacement-time graph in different situations,<br>e) draw a velocity-time graph from given | The learner is guided to:<br>- demonstrate motion by moving an object from a point to another for a specified distance and direction and state the meaning of the terms distance, displacement, speed, velocity and acceleration,<br>- participate in a race of a specified distance. Record time taken by each learner and work out the velocity and acceleration of different learners,<br>- work out the velocity and acceleration of movement of objects from one point to another, | 1. What is speed?<br>2. How do we use displacement-time graphs in real life?<br>3. How do we use velocity-time graphs? |

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| | - *Relative speed* | tables in different situations,<br>f) interpret velocity-time graph in different situations,<br>g) determine relative speed of two moving bodies in different situations,<br>h) appreciate use of linear motion in real-life situations. | - use data of displacement and time to plot displacement-time graphs on a grid or Cartesian plane,<br>- obtain information from a graph and examine it in relation to displacement, time and velocity,<br>- use data of velocity and time taken to draw a velocity-time graph on a grid or Cartesian plane,<br>- obtain information from a graph and examine it in relation to velocity, time and acceleration,<br>- observe two vehicles when one is overtaking, discuss and work out the relative speed of the two vehicles. Also discuss the dangers of careless overtaking<br>- use digital devices and other resources to explore more on linear motion in day-to-day life. | |

**Core Competencies to be developed:**
- Communication and Collaboration: the learner discusses different displacement-time graphs.
- Self-Efficacy: the learner participates in a race of a specified distance, record time taken by a learner and work out the velocity and acceleration.
- Critical Thinking and Problem Solving: the learner observes two bodies in motion such as cars when overtaking or moving in opposite directions and works out the relative speed of the two bodies in motion.

**Values:**
- Responsibility: as the learner uses objects in motion such as thrown objects and many others to explain distance, displacement, speed, velocity and acceleration.
- Integrity: as the learner participates in a race of a specified distance, records time taken by each learner and accurately works out the velocity and acceleration of different learners.
- Unity: as the learner participates and collaborates in a race to record time taken by each learner and work out the velocity and acceleration of different learners.

**Pertinent and contemporary Issues (PCIs)**:
- Learner Support Programmes (Sports and Games): the learner participates in a race of a specified distance, records time taken by each learner and works out the velocity and acceleration,
- Socio-Economic and Environmental Issues (Safety and Security): the learner observes the two bodies in motion such as cars when overtaking or moving in opposite directions and works out the relative speed of these two bodies in motion.
- Road safety: as the learner observes two vehicles when one is overtaking, discusses and works out the relative speed of the two vehicles. Also discusses the dangers of careless overtaking.

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**Suggested Assessment Rubric: Measurements and Geometry**

| Level<br>Indicators | Exceeds Expectations | Meets Expectations | Approaches Expectations | Below Expectations |
|---|---|---|---|---|
| Ability to determine the centre of enlargement and the linear scale factor from similar figures | The learner systematically and accurately determines the centre of enlargement and linear scale factor from similar figures | The learner accurately determines the centre of enlargement and linear scale factor from similar figures | The learner needs assistance to accurately determine the centre of enlargement and linear scale factor from similar figures | The learner partially determines the centre of enlargement or linear scale factor from similar figures |
| Ability to construct the image of an object under an enlargement given the centre and the linear scale factor | The learner accurately and precisely constructs objects and their images under enlargement given the centre and the linear scale factor | The learner accurately constructs objects and their images under enlargement given the centre and the linear scale factor | The learner partially accurately constructs objects and their images under enlargement given the centre and the linear scale factor | The learner partially constructs objects or their images under enlargement given the centre and the linear scale factor |
| Ability to carry out enlargement on the Cartesian plane | The learner accurately and systematically carries out enlargement on the Cartesian plane | The learner accurately carries out enlargement on the Cartesian plane | The learner partially accurately carries out enlargement on the Cartesian plane partly accurately | The learner partially carries out enlargement on the Cartesian plane |

| Level<br>Indicators | Exceeds Expectations | Meets Expectations | Approaches Expectations | Below Expectations |
|---|---|---|---|---|
| Ability to determine linear, area and volume scale factors | The learner precisely and correctly determines linear, area and volume scale factors | The learner correctly determines linear, area and volume scale factors | The learner correctly determines either linear or area or volume scale factors | The learner determines linear or area scale factors |
| Ability to apply similarity and enlargement to real life situations | The learner systematically and correctly applies similarity and enlargement to real-life situations | The learner correctly applies similarity and enlargement to real-life situations | The learner needs assistance to correctly partially correctly applies similarity and enlargement to some real life situations | The learner partially applies similarity and enlargement to real life situations |
| Ability to identify lines of symmetry of plane figures and properties of reflection | The learner precisely and correctly identifies lines of symmetry of plane figures and properties of reflection | The learner correctly identifies lines of symmetry of plane figures and properties of reflection | The learner correctly identifies lines of symmetry of plane figures or properties of reflection | The learner partially identifies lines of symmetry of plane figures or properties of reflection |
| Ability to draw an image given an object and a mirror line on a plane surface and Cartesian plane and determine the equation of the mirror line | The learner systematically and accurately draws an image given an object and a mirror line on a plane surface and Cartesian plane and determines the equation of the mirror line | The learner accurately draws an image given an object and a mirror line on a plane surface and Cartesian plane and determines the equation of the mirror line | The learner accurately draws an image given an object and a mirror line on a plane surface or Cartesian plane or determines the equation of the mirror line | The learner partially draws an image given an object and a mirror line on the plane surface or the Cartesian plane or determines the equation of the mirror line |

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| Ability to determine the properties of rotation and rotate an object, given the centre and angle of rotation | The learner comprehensively and correctly determines the properties of rotation and rotates an object, given the centre and angle of rotation | The learner correctly determines the properties of rotation and rotates an object, given the centre and angle of rotation | The learner correctly determines the properties of rotation or rotates an object, given the centre and angle of rotation | The learner partially determines the properties of rotation or rotates an object, given the centre and angle of rotation |
| Ability to determine the order of rotational symmetry in solids and relate rotation and congruence | The learner comprehensively and correctly determines the order of rotational symmetry in solids and relates rotation and congruence | The learner correctly determines the order of rotational symmetry in solids and relates rotation and congruence | The learner correctly determines the order of rotational symmetry in solids or relates rotation and congruence | The learner partially determines the order of rotational symmetry in solids or relates rotation and congruence |
| Ability to determine the trigonometric ratios of acute angles from right-angled triangles, mathematical | The learner systematically and accurately determines the trigonometric ratios of acute angles from right-angled triangles, | The learner accurately determines the trigonometric ratios of acute angles from right-angled triangles, | The learner partially determines the trigonometric ratios of acute angles from right-angled triangles, mathematical tables, and calculators | The learner determines the trigonometric ratios of acute angles from right-angled triangles, or mathematical tables, or calculators with assistance |

| Level Indicators | Exceeds Expectations | Meets Expectations | Approaches Expectations | Below Expectations |
|---|---|---|---|---|
| tables, and calculators | mathematical tables, and calculators | mathematical tables, and calculators | | |
| Ability to relate the sines, cosines, of complementary angles and sines, cosines, and tangents of acute angles | The learner logically and accurately relates the sines, cosines, of complementary angles and sines, cosines, and tangents of acute angles | The learner accurately relates the sines, cosines, of complementary angles and sines, cosines, and tangents of acute angles | The learner accurately relates the sines, cosines, of complementary angles or sines, cosines, or tangents of acute angles | The learner partially relates the sines, cosines, of complementary angles or sines, cosines, and tangents of acute angles |
| Ability to determine trigonometric ratios of special angles $30^0$, $45^0$, $60^0$, and $90^0$ using triangles, | The learner systematically and accurately determines trigonometric ratios of special angles $30^0$, $45^0$, $60^0$, and $90^0$ using triangles | The learner accurately determines trigonometric ratios of special angles $30^0$, $45^0$, $60^0$, and $90^0$ using triangles | The learner accurately determines trigonometric ratios of any two of the special angles $30^0$, $45^0$, $60^0$, and $90^0$ using triangles, | The learner partially determines trigonometric ratios of any one of the special angles $30^0$, $45^0$, $60^0$, and $90^0$ using triangles |

| Level Indicators | Exceeds Expectations | Meets Expectations | Approaches Expectations | Below Expectations |
|---|---|---|---|---|
| Ability to apply trigonometric ratios in the solution of triangles, angles of elevation, and depression | The learner appropriately and accurately applies trigonometric ratios in the solution of triangles, angles of elevation, and depression | The learner accurately applies trigonometric ratios in the solution of triangles, angles of elevation, and depression | The learner accurately applies trigonometric ratios in any two of either the solution of triangles or angles of elevation or depression | The learner partially applies trigonometric ratios in any one of either solution of triangles or angles of elevation or depression |

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| Ability to determine the area of quadrilaterals, heptagons, octagons, and irregular polygons | The learner systematically and accurately determines the area of quadrilaterals, heptagons, octagons, and irregular polygons | The learner accurately determines the area of quadrilaterals, heptagons, octagons, and irregular polygons | The learner accurately determines the area of quadrilaterals, heptagons, octagons, or irregular polygons | The learner partially determines the area of quadrilaterals, heptagons, octagons or irregular polygons |
| Ability to determine the area of an annulus, sector, annular sector, and segment of a circle | The learner systematically and accurately determines the area of an annulus, sector, annular sector, and segment of a circle | The learner accurately determines the area of an annulus, sector, annular sector, and segment of a circle | The learner accurately determines the area of an annulus, or sector, annular sector, or segment of a circle | The learner partially determines the area of an annulus or sector or annular sector or segment of a circle |

| Level<br>Indicators | Exceeds Expectations | Meets Expectations | Approaches Expectations | Below Expectations |
|---|---|---|---|---|
| Ability to determine the area of a common region between two intersecting circles | The learner systematically and correctly determines the area of the common region between two intersecting circles | The learner correctly determines the area of the common region between two intersecting circles | The learner partially correctly determines the area of the common region between two intersecting circles | The learner partially determines the area of the common region between two intersecting circles |
| Ability to determine the surface area of prisms, pyramids, cones, frustums and spheres, and composite solids | The learner systematically and accurately determines the surface area of prisms, pyramids, cones, frustums, spheres, and composite solids | The learner accurately determines the surface area of prisms, pyramids, cones, frustums, spheres, and composite solids | The learner accurately determines the surface area of prisms, pyramids, cones, frustums, spheres, or composite solids | The learner partially determines the surface area of prisms, pyramids, cones, frustums, spheres, or composite solids. |
| Ability to determine the volume of prisms, pyramids, cones, frustums, spheres, and composite solids | The learner consistently and correctly determines the volume of prisms, pyramids, cones, frustums, spheres, and composite solids | The learner correctly determines the volume of prisms, pyramids, cones, frustums, spheres, and composite solids | The learner correctly determines the volume of prisms, pyramids, cones, frustums, spheres, or composite solids | The learner partially determines the volume of prisms, pyramids, cones, frustums, spheres, or composite solids |

| Level<br>Indicators | Exceeds Expectations | Meets Expectations | Approaches Expectations | Below Expectations |
|---|---|---|---|---|
| Ability to represent vectors geometrically, perform operations on vectors and determine the magnitude of vectors | The learner precisely and correctly represents vectors geometrically, performs operations on vectors and determines the magnitude of vectors | The leaner correctly represents vectors geometrically, performs operations on vectors and determines the magnitude of vectors | The leaner correctly represents vectors geometrically, performs operations on vectors or determines the magnitude of vectors | The learner partially represents vectors geometrically, performs operations on vectors or determines the magnitude of vectors |
| Ability to determine translation vector | The learner systematically and accurately determines translation vector | The learner accurately determines translation vector | The learner partially accurately determines translation vector | The learner partially determines translation vector |
| Ability to determine displacement, speed, velocity, acceleration; | The learner explicitly and correctly determines displacement, speed, velocity, and acceleration | The learner correctly determines displacement, speed, velocity, and acceleration | The learner correctly determines displacement or speed or velocity or acceleration | The learner partially determines displacement or speed or velocity or acceleration |
| Ability to draw and interpret graphs of linear motion | The learner systematically and accurately draws and interprets graphs of linear motion | The learner accurately draws and interprets graphs of linear motion | The learner accurately draws or interprets graphs of linear motion | The learner partially draws or interprets graphs of linear motion |

| Level Indicators | Exceeds Expectations | Meets Expectations | Approaches Expectations | Below Expectations |
|---|---|---|---|---|
| Ability to solve problems involving relative speed | The learner correctly and systematically solves problems involving relative speed | The learner correctly solves problems involving relative speed | The learner partially correctly solves problems involving relative speed | The learner partially solves problems involving relative speed |

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**STRAND 3.0: STATISTICS AND PROBABILITY**
**Sub Strand 1: Statistics and Probability**

| Strand | Sub-Strand | Specific Learning Outcomes | Suggested Learning Experiences | Suggested Key Inquiry Question(s) |
|---|---|---|---|---|
| **3.0 Statistics and Probability** | **3.1. Statistics I**<br>**(16 lessons lessons)**<br>• *Collection of Data*<br>• *Frequency distribution table*<br>• *Mean, Mode and Median of grouped and ungrouped data*<br>• *Representation of Data*<br>• *Interpretation of Data* | By the end of the sub-strand the learner should be able to:<br>a) collect data from real-life sources,<br>b) draw a frequency distribution table for grouped and ungrouped data,<br>c) determine mean, mode and median of grouped and ungrouped data,<br>d) represent data using histograms and frequency polygons,<br>e) interpret data from histograms and frequency polygons,<br>f) promote data collection, | The learner is guided to:<br>• discuss with peers the meaning of the term data. Explore data sources and use different and appropriate methods to collect and record data,<br>• collect data on availability of pedestrian crossing points along major roads in the locality or collect any other data and use it to prepare a frequency distribution table for an ungrouped data,<br>• group collected data or any other data into appropriate classes and generate a frequency distribution table,<br>• discuss in pairs and identify the modal class for grouped data, | 1. What is statistics?<br>2. How do we represent data?<br>3. How do we use statistics in day-to-day life? |

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|---|---|---|---|---|
| | | organisation and representation for informed decision making. | • use data collected from different sources and work out the mode, median and mean for grouped and ungrouped data,<br>• work in a group and use collected data to draw histograms (*with equal and unequal class-width)* and frequency polygons,<br>• use the histograms and frequency polygons to examine data to make conclusions,<br>**Project**<br>Identify a situation or problem in the school or community. Collect data on it, organise the data and represent it using any methods to guide the school or community to make informed decisions. | |

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**Core Competencies to be developed:**

- Critical thinking and problem solving: the learner uses the histograms and frequency polygons to examine data and make conclusions.
- Self-Efficacy: the learner discusses with peers the meaning of the term data and explores data sources.
- Digital literacy: the learner uses digital devices to draw histograms and frequency polygons to make conclusions.
- Communication and collaboration**:** learner presents and shares findings from data to different groups of people such as school administrators, parents or local community leaders.

**Values**:

- Integrity: the learner collects and records data accurately.
- Unity: the learner works in groups and uses collected data to draw histograms and frequency polygons.
- Love: the learner is considerate of the other as they discuss in pairs and identify the modal class for grouped data

**Pertinent and Contemporary Issues (PCIs):**

- Life Skills (Decision Making): the learner makes informed decisions from data interpretation from histograms and frequency polygons.
- Parental Engagement and Empowerment: the learner seeks support and guidance from parents or guardians to collect, organise and present data on an identified situation in the school or community.

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**Sub-Strand: Probability 1**

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| Strand | Sub-Strand | Specific Learning Outcomes | Suggested Learning Experiences | Suggested Key Inquiry Question(s) |
|---|---|---|---|---|
| **3.0 Statistics and Probability** | **3.2 Probability 1**<br>**(12 lessons)**<br>- *Experimental probability*<br>- *Probability space*<br>- *Mutually exclusive and independent events*<br>- *Laws of Probability*<br>- *Probability tree diagrams* | By the end of the sub-strand the learner should be able to:<br>a) perform experiments involving probabilities in different situations,<br>b) identify the range of probability in different situations,<br>c) generate probability space of different events,<br>d) determine the probability of mutually exclusive and independent events,<br>e) apply the laws of probability in different situations,<br>f) determine the probability of independent events using tree diagrams,<br>g) appreciate the application of probability in real-life situations. | The learner is guided to:<br>- carry out experiments such as tossing coins or throwing dice to illustrate experimental probability,<br>- discuss with peers the probabilities of events and show the range of probability,<br>- list down all the possible outcomes of different events in real-life situations such as getting pedestrians crossing the roads at undesignated areas and highlight its dangers for public awareness,<br>- work out the probability of mutually exclusive events and independent events,<br>- work out probability of events using addition and multiplication law<br>- in pairs, use tree diagram to work out probability of two independent and mutually events,<br>- discuss, create and play games related to probability,<br>- use digital devices and other resources to learn more on probability. | How is probability applied in real-life situations? |

**Core** competencies to be developed:

- Communication and collaboration: the learner carries out experiments such as tossing coins or throwing dice to illustrate experimental probability.
- Citizenship: the learner acknowledges Kenyan currency as they carry out experiments of tossing coins.
- Self-efficacy: the learner appreciates his/her ability to play games related to probability.

**Values:**

- Responsibility: the learner carefully tosses coins or throws dice to illustrate experimental probability.
- Social justice: the learner plays games where they accept there is winning or losing and acknowledging honest outcomes.

**Pertinent and Contemporary Issues (PCIs):**

- Social Cohesion: the learner interacts with others freely as they discuss, create and play games related to probability.
- Life skills and values education: the learner uses different experimental events and lists down all the possible outcomes which assist in decision making.
- Road safety: the learner discusses with peers the probability of getting pedestrians crossing the roads at undesignated areas and highlighting its dangers for public awareness.

**Suggested Assessment Rubric: Statistics and Probability**

| Level / Indicators | Exceeds Expectation | Meets Expectation | Approaches Expectation | Below Expectation |
|---|---|---|---|---|
| Ability to draw a frequency distribution table for grouped and ungrouped data and determine mean, mode, and median | The learner accurately and systematically draws a frequency distribution table for grouped and ungrouped data and determines mean, mode, and median | The learner accurately draws a frequency distribution table for grouped and ungrouped data and determines mean, mode, and median | The learner needs assistance to accurately draw a frequency distribution table for grouped and ungrouped data or determines mean, mode, and median | The learner partially draws a frequency distribution table for grouped or ungrouped data or determines mean, mode or median |
| Ability to represent and interpret grouped data using histograms and frequency polygons | The learner systematically and accurately represents and interprets grouped data using histograms and frequency polygons | The learner accurately represents and interprets grouped data using histograms and frequency polygons | The learner accurately represents or interprets grouped data using histograms or frequency polygons | The learner partially represents or interprets grouped data using histograms or frequency polygons |
| Ability to perform experiments involving probabilities and identify the range of probability | The learner systematically and accurately performs experiments involving probabilities and identifies the range of probability | The learner accurately performs experiments involving probabilities and identifies the range of probability | The learner accurately performs experiments involving probabilities or identifies the range of probability | The learner partially performs experiments involving probabilities or identifies the range of probability |

| Level / Indicators | Exceeds Expectation | Meets Expectation | Approaches Expectation | Below Expectation |
|---|---|---|---|---|